Every run

Idle, 60 s	Warm-up: up to 4 requests	**Metered window: at least 60 s and 20,000 tokens, then requests in flight finish**	Idle, 60 s

Energy = the GPU's energy counter at the end of the window minus at its start, summed over the GPUs, then divided by the window's tokens. Most configurations ran twice, on two pods where two were available.

A decode request

cached prefix: 1,024 tokens | fresh: 200 → output: 256 tokens

The prefix is cached before the window; the fresh input is read inside it.

The ten experiments

c0 Loaded idle

What does a GPU draw while holding a model and doing nothing?

Nothing varied: a fixed 60 s window

16 runs · Section 4.1

c1 Batch

How does energy per token change with the batch?

Target batch 1, 4, 16, 64, 256; 1k cached, 200 fresh, 256 out

80 runs · Sections 4.2, 4.3

c2 Context

How does it change with the context read back?

Cached context 1k, 4k, 16k, model maximum; target batch 8, 64

141 runs · Section 4.4

c3 Prefill, cache off

What does reading a prompt cost per token?

Fresh prompt 1k, 4k, 16k, model maximum; target batch 1, 8; one output token

144 runs · Section 4.5

c4 Cached prefix

How much does reusing a cached prompt save?

Cached 1k, 4k, 16k; fresh 200 or 2,048; batch 1, 8; against an all-fresh control of equal length

384 runs · Section 4.6

c6 Tensor parallelism

What does splitting a model over more GPUs cost?

1 to 2 GPUs (Qwen3-8B), 2 to 4 (Llama-3.3-70B); batch 1, 16, 64

48 runs · Sections 4.3, 4.7

c7 Weight precision

What do 4-bit weights save against 8-bit?

fp8 or nvfp4; batch 8, 64; context 1k, 16k; B200 only

48 runs · Section 4.8

c8 Simulated agent

What does a workload that re-reads a growing context cost?

One agent, or 5 to 8 together; context from 10,240 tokens, 2,048 more per step

36 runs · Section 4.9

c9, c9b KV-cache pressure

What happens past the memory for stored context?

Batch 12 to 32 at 16k context; memory offered 90% or 70%; Llama-3.3-70B on B200

20 runs · Section 4.10